UNIT 6 · TOPIC 3

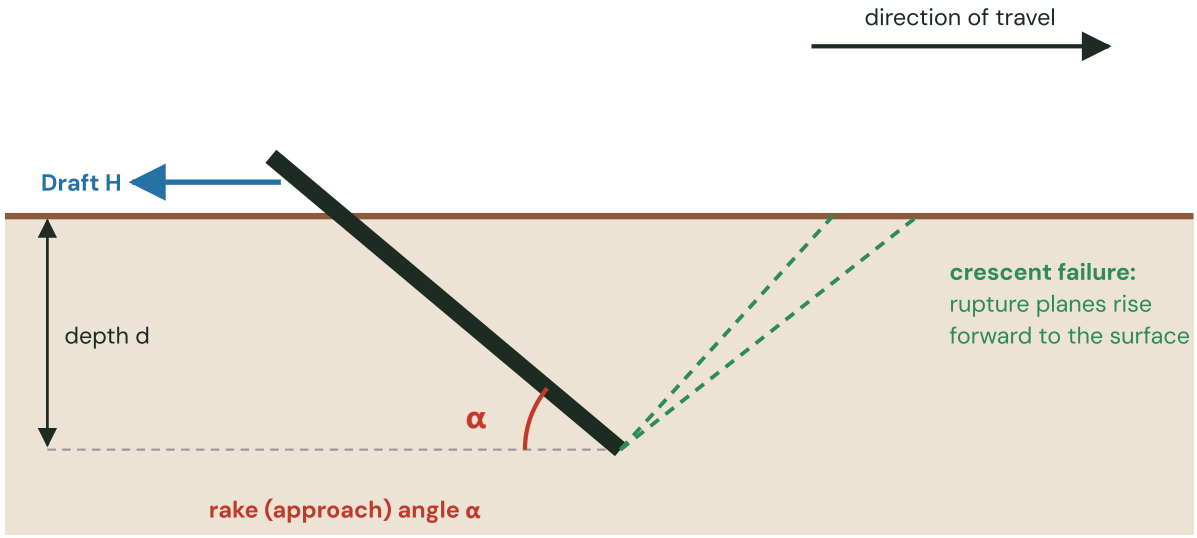
Mechanics of Tillage Tools

Syllabus topic: Mechanics of tillage tools

How a tillage tool works on soil

- Nearly all tillage machines apply pressure to the soil through INCLINED PLANES or WEDGES. As the implement advances, the soil ahead is compressed, and in an uncemented soil this produces a SHEARING action (Kepner §7.6, p. 143).
- So a tillage tool is a moving wedge: it compresses soil until it fails in shear (Mohr–Coulomb, Topic 2), then lifts, inverts, throws or pulverises the failed mass.
- Soil shear differs from shearing a solid: the reaction can spread a considerable distance either side of the shear plane, because of moisture-film cohesion and particle interlocking (internal friction) (p. 143).
- **Nichols (plastic soil):** shear force at a given pressure rises with moisture up to the LOWER PLASTIC LIMIT, then falls to zero at the UPPER (liquid) plastic limit — the physical basis of the friable-moisture tillage rule (Topic 2).

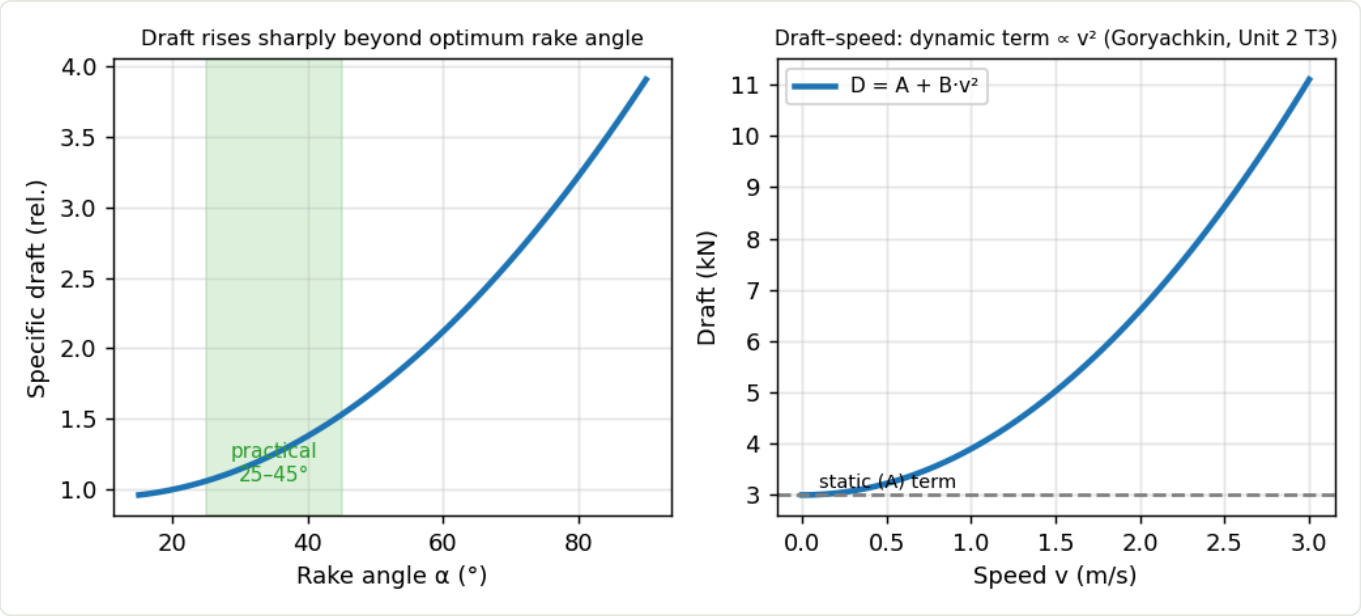
Rake (approach) angle and geometry



Narrow tine: forward crescent failure; below the critical depth → lateral compaction, no useful loosening

Mechanics of a soil-working tine (redrawn from the source figure)

- **RAKE ANGLE α** — the angle the working face makes with the horizontal — is the master geometric variable.
- Small α slices with low draft but little lift/inversion; large α lifts and pulverises more, but draft rises steeply beyond an optimum.
- Practical narrow tools use **~25–45°**; sweeps/blades use small angles for low-disturbance cutting.
- The wedge (bevel) angle and the width and depth of cut complete the geometry.



Draft rises sharply beyond the optimum rake angle; dynamic draft term ∝ v² (Goryachkin)

Soil failure ahead of the tool

Tool	Failure
NARROW tine (small width-to-depth ratio)	Forward CRESCENT: rupture surfaces rise from the tip forward and sideways to the surface, throwing blocks up and out — clean, low-energy failure, good tilth
Narrow tine below the CRITICAL DEPTH	No rupture to the surface — soil is compacted sideways/forward (lateral flow); draft rises without useful loosening and a compacted zone forms
WIDE blade	Behaves like a retaining wall — passive earth-pressure (Rankine/Terzaghi) analysis

The universal earthmoving (REECE) equation sums the draft contributions of soil weight, cohesion, surcharge and adhesion.

Force components and draft

- The total soil reaction resolves into draft (horizontal, along travel), vertical and side components (Unit 2, Topic 5).
- **Draft depends on:** soil strength (c, φ, cone index — Topic 1), tool geometry (α, width, depth), depth × width of cut, forward speed (dynamic term ∝ **v²**, Goryachkin), and soil–metal friction/adhesion.
- **Unit (specific) draft** = draft ÷ furrow cross-section (kN per cm², or kN/m of width) — the size-independent comparison (Unit 2, Topic 5; Unit 3, Topic 2b).

Soil–metal friction, adhesion and scouring (Kepner §7.5–7.7)

- Soil–metal friction governs both the LIFE and the DRAFT of a tillage tool (Kepner §7.5, p. 139).
- **Sliding-friction coefficient μ (soil–steel):** for a plastic soil μ rises with moisture; it is roughly independent of colloidal (clay) content above about **C = 32%**, and essentially independent of moisture in the pure "friction phase".
- Highly polished steel cut μ by **5–20%** for high-colloid soils, but not for sand (p. 143).
- **ADHESION** (soil sticking) causes poor SCOURING — sticky soil rides on the surface instead of shedding, raising draft and spoiling work; worst in wet high-clay (Indian black cotton) soils.

- **Remedies:** smooth/hard polished surfaces, enamel/PTFE coatings, the correct approach angle, and sometimes vibratory or powered tools.

Wear and abrasion

- Wear blunts edges and destroys geometry: shares/points lose SUCTION and penetration, and draft climbs.
- **Design responses:** high-carbon/boron steels, hard-facing at wear edges, replaceable/reversible points. Kepner §7.7 notes the shift from resharpened shares to DISPOSABLE shares (p. 145).

Practice MCQs

Q36. Fundamentally, a tillage tool acts on soil as a moving:

(a) Roller
(b) Inclined plane / wedge applying compressive stress
(c) Magnet
(d) Heater

Q37. In an uncemented soil, the compressive stress from a tool results in:

(a) Melting
(b) A shearing (failure) action
(c) Evaporation
(d) No change

Q38. Compared with shearing a solid, soil shear reaction:

(a) Is confined to one plane
(b) Extends some distance either side of the shear plane
(c) Is zero
(d) Is instant

Q39. The master geometric variable of a soil-working tool is the:

(a) Colour
(b) Rake (approach) angle
(c) Weight
(d) Serial number

Q40. As rake angle increases well beyond the optimum, draft:

(a) Falls
(b) Rises steeply
(c) Stays constant
(d) Becomes zero

Q41. A narrow tine fails soil ahead of it by:

(a) Melting
(b) A forward/upward crescent shear
(c) Compression only
(d) Boiling

Q42. Below the CRITICAL DEPTH, a narrow tine mainly causes:

(a) Crescent failure
(b) Lateral compaction with little useful loosening
(c) Inversion
(d) Pulverisation

Q43. Working a narrow tine too deep is inefficient because:

(a) Draft rises without useful loosening below the critical depth
(b) It is illegal
(c) Soil gets wetter
(d) Speed drops

Q44. A wide blade's soil resistance is analysed like a:

(a) Beam
(b) Retaining wall (passive earth pressure)
(c) Spring
(d) Pulley

Q45. The universal earthmoving (Reece) equation sums draft terms for:

(a) Fuel and speed
(b) Soil weight, cohesion, surcharge and adhesion
(c) Slip and sinkage
(d) Noise

Q46. Draft of a tillage tool increases with all EXCEPT:

(a) Greater depth and width
(b) Higher soil strength
(c) Lower rake angle within limits
(d) Higher speed

Q47. The dynamic (speed) component of tillage draft varies with:

(a) v
(b) v^2
(c) $\sqrt{v}$
(d) 1/v

Q48. Unit (specific) draft is draft per unit:

(a) Weight
(b) Furrow cross-sectional area
(c) Time
(d) Wheel

Q49. Nichols found plastic-soil shear force is maximum near the:

(a) Upper plastic (liquid) limit
(b) Lower plastic limit
(c) Oven-dry state
(d) Saturation

Q50. At the upper plastic limit the shear value falls to:

(a) Maximum
(b) Zero
(c) Unchanged
(d) Infinity

Q51. Soil–metal sliding friction μ is roughly independent of clay (colloidal) content above about:

(a) C = 5%
(b) C = 32%
(c) C = 60%
(d) C = 90%

Q52. Highly polished steel reduced μ by 5–20% for:

(a) Sand
(b) High-colloid (clay) soils
(c) Gravel
(d) All equally

Q53. Both the LIFE and the DRAFT of a tillage tool depend strongly on:

(a) Paint colour
(b) Soil–metal friction
(c) Tyre pressure
(d) Engine oil

Q54. Soil sticking to the tool surface (poor scouring) is due to:

(a) Cohesion only
(b) Adhesion, worst in wet high-clay soils
(c) Internal friction only
(d) Cone index

Q55. A remedy for poor scouring in sticky soils is:

(a) Rougher surface
(b) Polished/coated (enamel, PTFE) surface
(c) More weight
(d) Slower speed only

Q56. Wear/abrasion of a share primarily causes:

(a) Loss of suction and higher draft
(b) Lower draft
(c) Better inversion
(d) More suction

Q57. The modern trend in shares is toward:

(a) Welded permanent shares
(b) Disposable/replaceable shares
(c) Wooden shares
(d) No shares

Answers

Q36 — (b) Inclined plane / wedge applying compressive stress — *Kepner §7.6, p. 143 — tools as inclined planes/wedges causing shear.*

Q37 — (b) A shearing (failure) action

Q38 — (b) Extends some distance either side of the shear plane — *p. 143 — the shear reaction spreads either side of the plane.*

Q39 — (b) Rake (approach) angle

Q40 — (b) Rises steeply

Q41 — (b) A forward/upward crescent shear

Q42 — (b) Lateral compaction with little useful loosening — *Critical-depth concept (Gill & Vanden Berg).*

Q43 — (a) Draft rises without useful loosening below the critical depth — *Below the critical depth narrow-tine soil compacts laterally instead of rupturing to the surface.*

Q44 — (b) Retaining wall (passive earth pressure)

Q45 — (b) Soil weight, cohesion, surcharge and adhesion

Q46 — (c) Lower rake angle within limits — *A lower rake angle (within limits) REDUCES draft — the exception.*

Q47 — (b) v^2

Q48 — (b) Furrow cross-sectional area

Q49 — (b) Lower plastic limit — *Nichols' data in Kepner pp. 143–144 — shear peaks at the lower plastic limit.*

Q50 — (b) Zero — *Zero at the upper (liquid) plastic limit.*

Q51 — (b) C = 32% — *μ independent of colloid content above C ≈ 32%.*

Q52 — (b) High-colloid (clay) soils — *Polishing cuts μ 5–20% for high-clay soils, not sand.*

Q53 — (b) Soil–metal friction — *Kepner §7.5 — friction sets tool life and draft.*

Q54 — (b) Adhesion, worst in wet high-clay soils

Q55 — (b) Polished/coated (enamel, PTFE) surface

Q56 — (a) Loss of suction and higher draft — *Wear loses suction and raises draft.*

Q57 — (b) Disposable/replaceable shares — *Kepner p. 145 — trend to disposable shares.*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 6–7 (force analysis; moldboard tools), pp. 116–169; Gill & Vanden Berg — USDA Agriculture Handbook 316; Nichols' soil–metal friction/shear studies (cited in Kepner); Reece universal earthmoving equation; O'Callaghan & Payne tine mechanics; NPTEL soil-dynamics lectures.